Lecture #12 / 15 October 2026

PS #2 due Friday @ 11:59 PM (electronic)
PS #3 due Tuesday 10/27
PS #4 due Thursday 11/5
Midterm Exam 11/5

Office Hours: Monday @ 2:30 PM (will send out a link)

Review: **PID**

$$u(t) = K_p e + K_I \int_0^t e(\sigma)\,d\sigma + K_D \dot{e}$$

$$U(s) = \left[\frac{(K_D s^2 + K_p s + K_I)}{s}\right] E(s)$$

→ A PID controller introduces 2 zeros and one pole onto the open loop transfer function

Example:

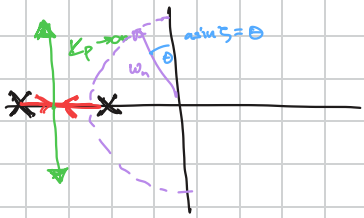
2nd Order Plant under P-control

Plant: $G(s) = \dfrac{A}{s^2 + a_1 s + a_2}$ ⟹ Char. Poly for CL was $s^2 + a_1 s + a_2 + K_p A = 0$

- Proportional Control affected the pole locations of the closed-loop dynamics
- " " the plant kept its original order
- CL char. poly. as $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$

$$\omega_n^2 = a_2 + K_p A \quad | \quad 2\zeta\omega_n = a_1$$

K_p changed both ω_n and ζ



Steady State Error (SSE) under P control (with respect to step)

$$\frac{E}{R} \quad \text{or} \quad \frac{Y}{R}$$

$$\frac{E}{R} = \frac{1}{1 + K_p G} = \frac{1}{1 + K_p\left(\frac{A}{s^2 + a_1 s + a_2}\right)} = \boxed{\frac{s^2 + a_1 s + a_2}{s^2 + a_1 s + a_2 + K_p}}$$

Step ⇒ $R = \frac{1}{s}$ ⇒ $E = \frac{1}{s}$

SSE → FVT ⇒ $\lim_{t\to\infty} e(t) = \lim_{s\to 0} s \cdot E(s) = \lim_{s\to 0} s \cdot \frac{1}{s} \cdot$ (boxed above) $= \boxed{\frac{a_2}{a_2 + K_p}}$

nonzero steady state error

$K_p \to \infty$ ⇒ $e_{ss} \to 0$

Add integral Control to eliminate / reduce SSE

(tradeoff for pure proportional: $K_p \to \infty$, $e_{ss} \to 0$, $\zeta \to 0$)

Integrators have "memory" (accumulators)

↳ They want to drive accumulated error to ∅

and consequently, that (usually) drives the error itself to ∅

Ex: $\frac{E}{R} = \frac{1}{1+KG} = \frac{1}{1+\frac{K}{s}\cdot G} = \frac{s}{s+K_1 G}$ $\left(R_i = \frac{1}{s}\right)$

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} s\cdot E(s) = \lim_{s\to 0} s\cdot \left(\frac{1}{s}\right)\left(\frac{s}{s+K_1 G(s)}\right)$$

$$= \lim_{s\to 0} \frac{s}{s+K_1 G(s)} = \frac{0}{K_1 G(0)}$$

if $G(0) > 0$ (but also finite) then

$e_{ss} = \emptyset$

K_p had nonzero SSE, but when used an integrator the SSE was zero (to a step input)

Derivative Control

* Never alone, always with PD or PID

* PD is better b/c $K_p e(t) + K_D \dot{e}(t) = K_p\left(e(t) + T_D \dot{e}(t)\right)$

$$\approx K_p e(t+T_D)$$

predicting forward in time

"Adds phase"



Prop. / Integral: $u(s) = \frac{K_P s + K_I}{s}$ → one zero and one pole that we add to the open loop system

Ex: $G(s) = \frac{K_0}{(\tau_1 s+1)(\tau_2 s+1)}$ $\tau_1 = 1s$, $\tau_2 = 10s$, $K_0 = 1000$

↑ limiting factor (slow)

Goals for Closed Loop: negligible overshoot and robustness to modeling errors in the params K_0, τ_1, τ_2

Quick Aside: what is robustness?

$G(s)$ — but it could actually be this

but what we assume is $\tilde{G}(s)$ — design our control law for this

want stability and performance to be "good" in light of plant uncertainty

Case #1: Open Loop → $u = 300/K_0 = 0.3$

it settles to 1% in 47.1 sec. → why? b/c of the "slow pole" in the plant TF

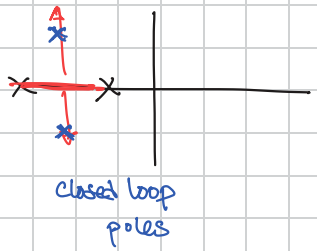
Case #2: P Control (K_P) → $u = 0.03 \cdot e(t)$

⇒ char poly: $(s+1)(10s+1)+30 = 0$

$10s^2 + 11s + 31 = 0$

$\omega_n = 1.761$ rad/s $\quad \zeta = 0.312$ → poor damping

$t_r \approx \frac{1.8}{\omega_n} \quad t_s \approx \frac{4.6}{\sigma}$



closed loop poles

PI Control: $K_p = 0.03$ $K_I = 0.003$

$$U(s) = \frac{(0.03)s + 0.003}{s} \cdot E(s)$$

$$K \cdot G = \left(\frac{0.03(s+0.1)}{s}\right)\left(\frac{100}{(s+1)(s+0.1)}\right) = \frac{3}{s(s+1)}$$

$$T(s) = \frac{Y}{R} = \frac{KG}{1+KG} = \frac{3}{s^2+s+3}$$

$$\Rightarrow \omega_n = 1.732 \text{ rad/s} \qquad \zeta = 0.289$$

Can conclude that for these specific gains, the performance (nominal) is roughly the same.

Q: Why do we care about P vs. PI?

Didn't just (only) care about performance but also

ROBUSTNESS

Hypothesis: PI control is more robust to plant uncertainty than simple P control.

Answer: The integrator adds robustness.

We'll learn some tools that will help us analyze this later on.

(Teaser: integrals add higher gain at lower frequency)

Final: PID

$$G(s) = \frac{A}{s^2 + a_1 s + a_2}$$

$\Rightarrow$ char. poly. $\rightarrow$ $s^2 + a_1 s + a_2 + A\left(K_P + \frac{K_I}{s} + K_D s\right) = 0$

$\rightarrow$ $\boxed{s^3 + (a_1 + AK_D)s^2 + (a_2 + AK_P)s + AK_I = 0}$

This determines the pole locations for a 2nd order plant.

$\Rightarrow$ 3 closed loop poles

"Pole Placement" $\rightarrow$ we want to place the closed loop poles wherever we want.

Desired Char. Poly: $s^3 + b_1 s^2 + b_2 s + b_3 = 0$

(we choose b_1, b_2, b_3)

$$b_1 = a_1 + AK_D$$
$$b_2 = a_2 + AK_P$$
$$b_3 = AK_I$$

What do we choose for K_P, K_I, K_D

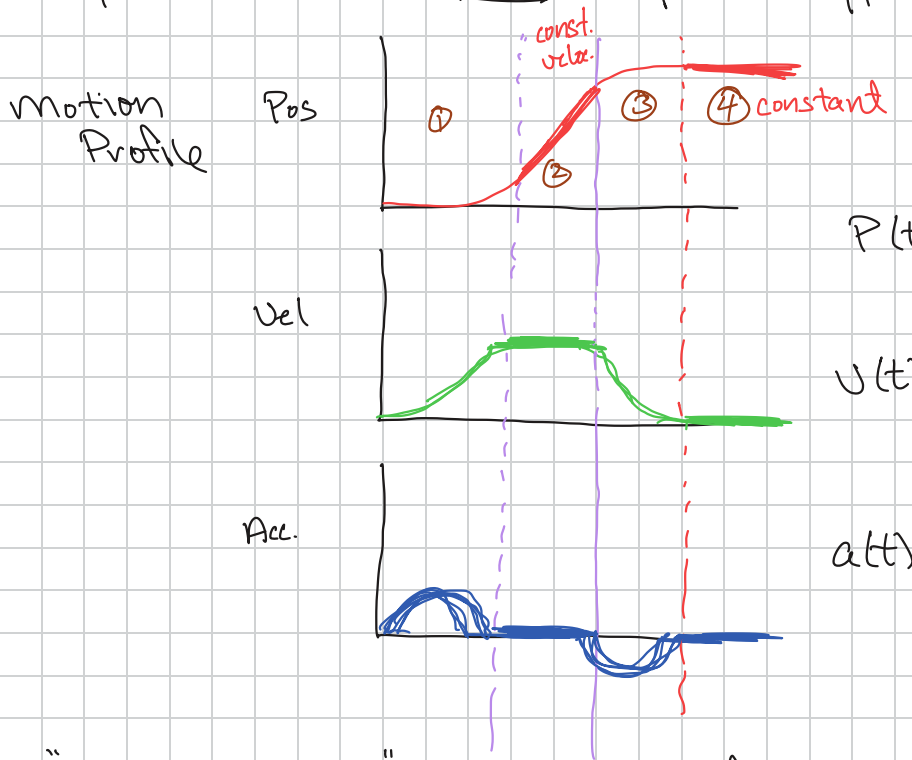
Answer: you just solve for them.

$\Rightarrow$ We can implement pole placement for a second order plant using PID.

Summarize PID

Increase	Rise Time	Overshoot	Settling Time	SSE to a Step	Stability Margin
K_P	decrease	increases	small change	decreases	usually decreases
K_I	decrease	increases	increases	eliminates	decrease
K_D	small change	decreases "add phase"	decreases	no effect	increase

Steady State Error and System Type



$P(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$

$v(t) = a_1 + 2a_2 t + 3a_3 t^2 + 4a_4 t^3 + 5a_5 t^4$

$a(t) = \cdots$

"System Type" is a proxy for how well a system can track a polynomial of a certain degree

(2) $P = P_0 + v_0 t$ ← polynomial

(other) $P = P_0 + v_0 t + at^2$ ← polynomial

Polynomial Inputs:

"causal"

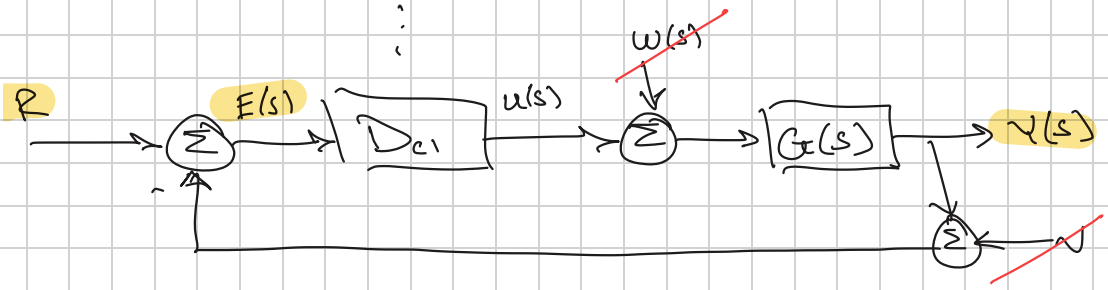
$$r(t) = \frac{t^k}{k!} 1(t) \Rightarrow R(s) = \frac{1}{s^{k+1}}$$

$k=0 \Rightarrow$ step

$k=1 \Rightarrow$ ramp

$k=2 \Rightarrow$ parabola

$\vdots$



$$E = S \cdot R = \frac{1}{1 + D_{c1} G} \cdot R \Rightarrow e_{ss} = \lim_{s \to 0} s \cdot E(s)$$

$$= \lim_{s \to 0} s \cdot R \frac{1}{1 + D_{c1} G}$$

$$R = \frac{1}{s^{k+1}}$$

$$= \lim_{s \to 0} \frac{1}{s^k} \cdot \frac{1}{1 + D_{c1} G}$$

Pull all integrators into a single term.

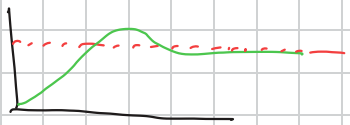
(We want to count the integrators)

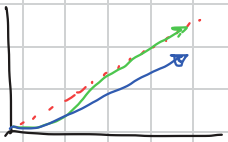
$$G D_{c1} = \frac{G D_{c0}}{s^n} \rightarrow G D_{c0}(0) = K_n \text{ finite and nonzero}$$

$$e_{ss} = \lim_{s \to 0} \frac{1}{1 + G D_{c0}(s)/s^n} \cdot \frac{1}{s^k}$$

$$= \lim_{s \to 0} \frac{s^n}{s^n + K_n} \cdot \frac{1}{s^k}$$

what is this with respect to k and n





$n > K$

$n = K$ (lag)

$e_{ss} =$

$$\begin{cases} 0 & n > K \\ \dfrac{1}{1+K_0} & n = K = 0 \\ \dfrac{1}{K_n} & n = K \geq 1 \\ \infty & n < K \end{cases}$$

n is the # of integrators in open loop

K is the order of polynomial

What is the intuition?

Integrators in the open loop system help you track higher order polynomials (w/ zero or finite SSE)

This integer n is the "type" of the system. For unity feedback loop, it's the # of poles of GD_{cl} at $s=0$ (aka the origin)

→ Counting the controller's integrators and the plant's integrators

A type "K" system follows a polynomial of degree "K" with nonzero constant error

$$K_p = \lim_{s \to 0} GD_{cl} \qquad K_v = \lim_{s \to 0} s \cdot GD_{cl} \qquad K_a = \lim_{s \to 0} s^2 \cdot GD_{cl}$$

position — velocity — acceleration

error constants

Type	(K=0) Step	(K=1) Ramp	(K=2) Parabola
0	$\frac{1}{1+K_p}$	∞	∞
1	0	$\frac{1}{K_v}$	∞
2	0	0	$\frac{1}{K_a}$

Steady State Error when tracking